Year 4 Mathematics

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Learning Strategies

Mathematics is often the most challenging subject for students. Much of the trouble comes from the fact that mathematics is about logical thinking, not memorizing rules or remembering formulas. It requires a different style of thinking than other subjects. The students who seem to be "naturally" good at math just happen to adopt the correct strategies of thinking that math requires – often they don't even realise it. We have isolated several key learning strategies used by successful maths students and have made icons to represent them. These icons are distributed throughout the book in order to remind students to adopt these necessary learning strategies:



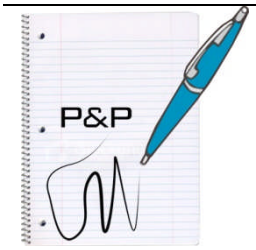
Talk Aloud Many students sit and try to do a problem in complete silence inside their heads. They think that solutions just pop into the heads of 'smart' people. You absolutely must learn to talk aloud and listen to yourself, literally to talk yourself through a problem. Successful students do this without realising. It helps to structure your thoughts while helping your tutor understand the way you think.



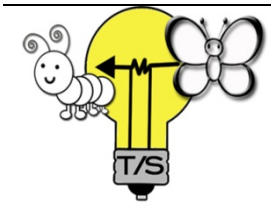
BackChecking This means that you will be doing every step of the question twice, as you work your way through the question to ensure no silly mistakes. For example with this question: $3 \times 2 - 5 \times 7$ you would do "3 times 2 is 5 ... let me check – no 3×2 is 6 ... minus 5 times 7 is minus 35 ... let me check ... minus 5×7 is minus 35. Initially, this may seem time-consuming, but once it is automatic, a great deal of time and marks will be saved.



Avoid Cosmetic Surgery Do not write over old answers since this often results in repeated mistakes or actually erasing the correct answer. When you make mistakes just put one line through the mistake rather than scribbling it out. This helps reduce silly mistakes and makes your work look cleaner and easier to backcheck.



Pen to Paper It is always wise to write things down as you work your way through a problem, in order to keep track of good ideas and to see concepts on paper instead of in your head. This makes it easier to work out the next step in the problem. Harder maths problems cannot be solved in your head alone – put your ideas on paper as soon as you have them – always!



Transfer Skills This strategy is more advanced. It is the skill of making up a simpler question and then transferring those ideas to a more complex question with which you are having difficulty.

For example if you can't remember how to do long addition because you can't recall exactly how to carry the one: $\begin{array}{r} 5889 \\ +4587 \\ \hline \end{array}$ then you may want to try adding numbers which you do know how to calculate that also involve carrying the one: $\begin{array}{r} 5 \\ +9 \\ \hline \end{array}$

This skill is particularly useful when you can't remember a basic arithmetic or algebraic rule, most of the time you should be able to work it out by creating a simpler version of the question.

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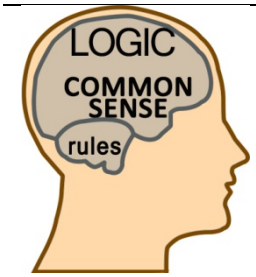
Format Skills These are the skills that keep a question together as an organized whole in terms of your working out on paper. An example of this is using the "=" sign correctly to keep a question lined up properly. In numerical calculations format skills help you to align the numbers correctly.

This skill is important because the correct working out will help you avoid careless mistakes. When your work is jumbled up all over the page it is hard for you to make sense of what belongs with what. Your "silly" mistakes would increase. Format skills also make it a lot easier for you to check over your work and to notice/correct any mistakes.

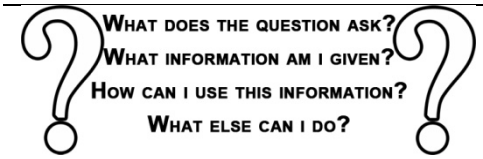
Every topic in math has a way of being written with correct formatting. You will be surprised how much smoother mathematics will be once you learn this skill. Whenever you are unsure you should always ask your tutor or teacher.



Its Ok To Be Wrong Mathematics is in many ways more of a skill than just knowledge. The main skill is problem solving and the only way this can be learned is by thinking hard and making mistakes on the way. As you gain confidence you will naturally worry less about making the mistakes and more about learning from them. Risk trying to solve problems that you are unsure of, this will improve your skill more than anything else. It's ok to be wrong – it is NOT ok to not try.



Avoid Rule Dependency Rules are secondary tools; common sense and logic are primary tools for problem solving and mathematics in general. Ultimately you must understand Why rules work the way they do. Without this you are likely to struggle with tricky problem solving and worded questions. Always rely on your logic and common sense first and on rules second, always ask Why?



Self Questioning This is what strong problem solvers do naturally when they get stuck on a problem or don't know what to do. Ask yourself these questions. They will help to jolt your thinking process; consider just one question at a time and Talk Aloud while putting Pen To Paper.

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Year 4 Mathematics

Number

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Useful formulae and hints

Numbers are written in the form “abcd”, where each letter represents a digit

d is the number of ones in the number

c is the number of tens in the number

b is the number of hundreds in the number

a is the number of thousands in the number

For example: the number 4325 has 4 thousands, 3 hundreds, 2 tens, and 5 ones. These are called the place values of the digits

To group numbers from largest to smallest, work from the left of the number. Compare all the three digit numbers first.

For example: comparing 4325, 4346, 4327, 137, 5401, and 153

Of the four digit numbers, there is only one with 5 thousands; that must be the biggest

If the thousands digit is the same, compare the hundreds digits

If the hundreds digits are the same, compare the tens digits

The next largest number is 4346

If numbers have the same hundreds and tens digits, compare their units’ digits.

4327 is bigger than 4325

Once all the three digit numbers have been compared, do the same for the three digit numbers; 153 is greater than 137

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Do the same for single digit numbers if there are any

To group smallest to largest, follow the above rules but start with the single digit numbers, then two digits, then three, then four

When deciding how to solve word problems, look for key words

More than, together means addition

Less than, difference means subtraction

Times means multiplication

Share means division

When looking for number patterns, work out the difference between two numbers next to each other. See if that rule works for the next two numbers. If it does, use your rule to complete the pattern

Fractions are in the form $\frac{Some\ number}{Some\ number}$

The bottom number is called the denominator and shows the total number of **equal parts** something is broken up into.

The top number is called the numerator, and shows how many of these parts we have

For example, the fraction $\frac{3}{4}$ shows that something is made up of four equal parts, and we have three of these parts

(Think of a cake or pizza)

To change a fraction to a decimal, divide the numerator by the denominator

To change a percentage to a decimal, remove the percentage sign an move the decimal point two places to the left

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To change a percentage to a fraction, remove the percentage sign, put the number as a fraction with 100 as the denominator, and simplify the fraction if necessary

When working out possible events, all possibilities must be listed and counted.

For example, if there are 3 children in a family there could be

3 girls

2 girls and a boy

2 boys and a girl

3 boys

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Exercise 1

Representing Numbers

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1) Write as numbers

a) Three hundred and ninety

b) Eight hundred and eighty three

c) Seven hundred and ninety three

d) Five hundred and six

e) Nine hundred and nine

2) Write as numbers

a) Two thousand two hundred and three

b) Seven thousand four hundred and ninety seven

c) Eight thousand six hundred and thirty

d) Nine thousand and twenty one

e) Three thousand and one

3) Write in words

a) 2713

b) 2097

c) 3330

d) 8090

e) 2010

f) 1117

g) 0

4) Write down the number that comes before each of these numbers

a) 331

b) 156

c) 905

d) 120

e) 1710

f) 1100

g) 2442

h) 1900

i) 9001

j) 3006

k) 1234

l) 10000

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5) Write the number that comes after each of these numbers

a) 819

b) 1090

c) 8881

d) 4223

e) 8010

f) 711

g) 1999

h) 3009

6) Put these numbers in order from smallest to largest

1325, 1101, 1123, 3000, 2946, 2121, 1015, 2221, 2323, 9104, 694

7) Put these numbers in order from largest to smallest.

2015, 2004, 4020, 1912, 1911, 2333, 3322, 2921, 2221, 4121, 3004

8) What is the value of the number 4 in each of these numbers?

a) 1034

b) 1435

c) 2114

d) 4027

e) 4

f) 1040

g) 2047

9) Use the > or < sign to show the relationship between the following pairs of numbers

a) 1234, 2134

b) 9821, 9281

c) 8005, 8015

d) 1023, 103

e) 970, 907

f) 1099, 1089

10) Write the number that is 10 less than the number shown. Repeat 4 times

a) 675

b) 555

c) 390

d) 442

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e) 530

f) 401

g) 112

h) 220

i) 1039

j) 1050

k) 908

11) Write the number that is 10 more than the number shown. Repeat four times

a) 1121

b) 2020

c) 3175

d) 1099

e) 803

f) 960

g) 999

h) 100

i) 1251

12) Round the following numbers to the nearest thousand, hundred and ten

a) 1263

b) 926

c) 101

d) 4565

e) 8555

f) 7550

g) 6005

h) 1111

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Exercise 2

Addition & Subtraction

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1) Add these numbers

a) 632 + 114

b) 247 + 319

c) 621 + 535

d) 877 + 223

e) 135 + 175

f) 414 + 441

2) Add these numbers

a) 2225 + 529

b) 4302 + 410

c) 8009 + 377

d) 4335 + 323

e) 8122 + 110

f) 9334 + 73

3) Subtract these numbers

a) 816 - 412

b) 594 - 482

c) 756 -511

d) 929 - 353

e) 504 - 127

f) 865 – 821

g) 9026 – 312

h) 6111 -- 3227

4) Peter has 840 stamps, John has 275 stamps. How many stamps do they have between them?

5) Alan weighs 145 kg, Chris weighs 148 kg. How much do they weigh together?

6) There were 1510 more people at the football game than at the rugby. If there were 4600 people at the football how many people were at the rugby?

7) Tom and Jerry have read 410 books between them. If Tom has read 318 books, how many books has Jerry read?

8) 138 students passed a test, 112 failed, and 35 were absent. How many students are in the school?

9) What number is 299 less than 6075?

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10) What is the difference between 2710 and 3244?

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Exercise 3

Multiplication & Division

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1) Calculate the following

a) 5×10

b) 5×20

c) 5×30

d) 40×5

e) 60×5

f) 20×7

g) 40×7

h) 60×7

i) 60×9

2) Calculate the following

a) 8×13

b) 16×9

c) 11×7

d) 17×8

e) 32×6

f) 45×9

3) Calculate the following

a) 15×6

b) 15×8

c) 6×15

d) 7×15

e) 9×15

f) From your answers, state a method for quickly multiplying any number by 15

4)

a) How many fours in 24?

b) What is 24×25?

c) How many fours in 28?

d) What is 28×25?

e) How many fours in 32?

f) What is 32×25?

g) Use your answers to parts a to f to state a method for quickly multiplying any number by 25

5) Calculate the following

a) $24 \div 5$

b) $33 \div 8$

c) $15 \div 4$

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d) $35 \div 7$

e) $24 \div 7$

f) $74 \div 7$

g) $37 \div 5$

h) $49 \div 8$

i) $21 \div 4$

j) $82 \div 8$

6) Write the factors of the following

a) 9

b) 15

c) 24

d) 7

e) 4

f) 1

g) 64

h) 100

i) 22

7) Mary has 40 lollies. If she gives each of her 6 friends an equal amount of lollies, how many will she have left over for herself? (She gives each friend the most that she can)

8) Alan buys 5 pens and gets 5 cents change from his dollar. How much was each pen?

9) Kathy is having a birthday party and wants each friend to get five lollies in their party bag. If there are 8 friends coming to the party, how many lollies will be left over from a bag of 50?

10) Tom has $5 left after giving an equal amount of money to a number of charities. If he started with $35, list how many charities he may have given money to, and how much he would have given to each.

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Exercise 4

Number Patterns

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1) Find the sixth term in the following sequences

a) 3, 6, 9, 12

b) 2, 4, 6

c) 5, 10, 15

d) 7, 14, 21

e) 4, 8, 12

f) 9, 18, 27

2) Find the fifth term in the following sequences

a) 25, 20, 15

b) 40, 32, 24

c) 63, 54, 45,

d) 63, 60, 57

e) 14, 11, 8, ___, ___

3) Find the missing numbers

a) $\square + 12 = 20$

b) $\square + 10 = 20$

c) $\square$ x $5 = 30$

d) 11 x $\square = 44$

e) $7 + \square = 15$

f) $\square$ x $3 = 21$

g) $\square + 10 = 15$

4) Complete the following sequences

a) $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}$, ____, _____

b) $\frac{1}{3}, \frac{2}{3}, 1$, ____, ____

c) $\frac{1}{5}, \frac{2}{5}, \frac{3}{5}$, ____, ____

d) $\frac{5}{3}, \frac{4}{3}, 1$, ____,____

e) $\frac{10}{7}, \frac{9}{7}, \frac{8}{7}$, _____,_____

f) $\frac{97}{100}, \frac{98}{100}, \frac{99}{100}$, _____,_____

5) Peter wants to give 8 people $5 each. If he has $32 how much more money does he need to be able to do this?

6) There are 9 tables in a restaurant. Each table has 6 chairs around them. If there are 70 people coming to the restaurant at one time, how many more chairs are needed?

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7) Every minute 5 ants crawl out of an ant hill.

a) How many ants have crawled out after 4 minutes?

b) There are 50 ants out of the ant hill. How many more minutes will go by until there are 75 ants out of the ant hill?

8) After 4 hours there were 24 cars in a car park. If the same number of cars park each hour

a) How many cars will be in the car park after 7 hours?

b) How many hours will have passed until there are 54 cars in the car park?

c) If the car park holds 96 cars, how long until it is full from when it first opened?

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Exercise 5

Fractions

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1) Write the following as a fraction

a) One fifth

b) One tenth

c) Two fifths

d) One hundredth

e) Three fifths

f) Three tenths

g) Seventeen hundredths

h) Four fifths

i) Nine tenths

2) Write the following in words

a) $\frac{1}{5}$

b) $\frac{1}{100}$

c) $\frac{3}{10}$

d) $\frac{11}{100}$

e) $\frac{7}{10}$

f) $\frac{4}{5}$

g) $\frac{99}{100}$

3) Put these fractions in order from smallest to largest

$\frac{3}{5}, \frac{2}{5}, \frac{4}{5}, \frac{1}{5}$

4) Put these fractions in order from largest to smallest

$\frac{5}{10}, \frac{1}{10}, \frac{7}{10}, \frac{2}{10}, \frac{6}{10}$

5) Fill in the missing numbers

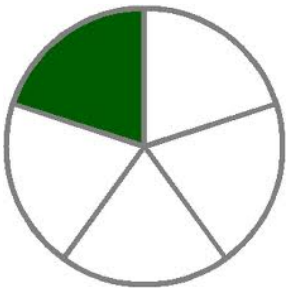
$\frac{97}{100}, \frac{95}{100}, \frac{93}{100}, \frac{91}{100}, __, __$

6) Fill in the missing numbers

$\frac{11}{5}, \frac{14}{5}, __, \frac{20}{5}, __, __$

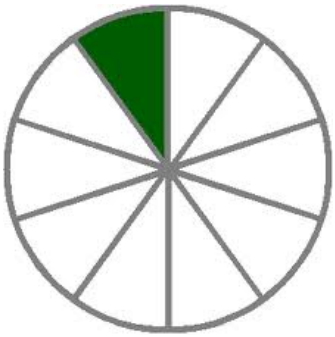
7) What fraction is shaded in the following diagrams?

a)

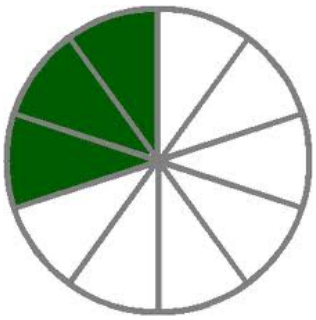


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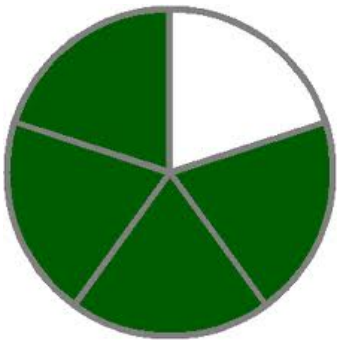
b)



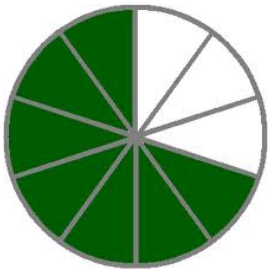
c)



d)



e)



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8) Place the fractions $\frac{1}{10}, \frac{1}{5}, \frac{20}{100}, \frac{2}{5}, \frac{7}{10}, \frac{75}{100}, \frac{4}{5}, \frac{95}{100}$ on a number line

9) Tim has one fifth of his lollies left, while Jack has eaten two fifths. Who has more lollies left?

10) Peter had $100 and spent $50. Jack had $10 and spent only $3. Who spent the bigger fraction of their money?

11) A fly spray kills two fifths of the flies in a room, whilst another kills three tenths of them. Which fly spray works better?

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Exercise 6

Decimals & Percentages

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1) Round the following decimals to the nearest whole number

a) 1.48

b) 11.05

c) 13.74

d) 0.22

e) 1.55

f) 22.51

2) Express the following fractions and mixed numbers as decimals

a) $\frac{3}{10}$

b) $\frac{15}{100}$

c) $3\frac{1}{10}$

d) $1\frac{7}{10}$

e) $1\frac{7}{100}$

f) $1\frac{77}{100}$

3) Multiply each of the following by 10

a) 1.4

b) 2.5

c) 3.7

d) 5.8

e) 10.2

f) 1.36

g) 2.45

h) 6.22

i) 8.49

j) 15.43

4) Multiply each of the following by 100

a) 1.52

b) 2.75

c) 4.26

d) 8.04

e) 13.11

f) 8.6

g) 7.2

h) 4.3

i) 1.2

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5) Write the following as a decimal

a) 30%

b) 15%

c) 20%

d) 10%

e) 75%

f) 90%

g) 100%

6) Write the following as a fraction

a) 50%

b) 25%

c) 10%

7) Divide each of the following by 10

a) 13.2

b) 10.8

c) 9.6

d) 7.2

e) 3.3

f) 1

8) Divide each of the following by 100

a) 152.5

b) 143.2

c) 131.9

d) 106.5

e) 98.9

f) 90.2

g) 66.6

h) 9.25

9) Alex has $14.25 in his bank account. Tom has ten times as much. How much money does Tom have?

10) John runs 30km and Jill runs 50% of that distance. How far did Jill run?

11) Place the following decimals on a number line

0.7, 0.65, 0.8, 0.1, 0.25, 0.4, 0.5, 0.9, 0.45

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12) Express the following as a decimal

a) $\frac{51}{1000}$

b) $\frac{74}{1000}$

c) $\frac{17}{1000}$

d) $\frac{7}{1000}$

e) $\frac{1}{1000}$

13) Calculate the following

a) 1.2 + 3.4

b) 3.6 + 4.3

c) 10.2 + 5.3

d) 1.25 + 3.1

e) 2.56 + 5.2

f) 7.4 + 2.22

g) 8.1 + 3.05

14) Calculate the following

a) $7.4 - 2.3$

b) $9.6 - 3.1$

c) $10.7 - 9.6$

d) $8.4 - 4.8$

e) $3.2 - 2.5$

f) $7.65 - 4.3$

g) $3.43 - 2.3$

h) $5.69 - 3.06$

i) $7.32 - 5.61$

j) $8.19 - 5.43$

15) Jake has $14.70 and spends $12.35. How much money does he have left?

16) Paul has $12.35 and his grandfather gives him $11.15. How much money does Paul now have?

17) Barbara wants to save up to buy a new dress that costs $35.30. At the moment she has $16.10. How much more money does she need to be able to buy the dress?

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Exercise 7

Chance

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1) Alan tosses two coins. List the possible combinations they could land on

2) Peter rolls two dice and adds the two numbers. List all the numbers that he could get

3) List what the two dice from question 2 could show to get a total of 7

4) List what the two dice from question 2 could show to get a total of 12

5) There are 6 red shirts, 6 blue shirts and 6 yellow shirts in a draw. If a boy pulls a shirt out without looking:

a) List what colour shirt he might pull out

b) Which colour shirt will he probably pull out?

c) Could he pull out 6 yellow shirts in a row?

6) There are 20 red, 20 blue and 20 green lollies in a jar. If Jack closes his eyes and chooses one:

a) What colour lolly will he probably choose?

b) What colour lolly could he not get?

c) If he pulls out a red lolly first time, will he definitely get a red lolly next time?

d) Could he pull out 20 red lollies in a row?

e) If he did this, which colour would he be more likely to pull out in his next turn?

7) In a jar there are 20 blue buttons. In another jar there are 20 blue and 20 yellow buttons.

a) Which jar has more blue buttons?

b) From which jar is he more likely to pull out a blue button?

c) Is he more likely to pull a yellow or blue button from the second jar?

d) Could he pull 20 yellow buttons in a row from the second jar

e) If he did this, from which jar would he then have more chance of pulling a blue button from?

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8) Tom rolls two normal 6 sided dice and adds the numbers. Which total is he most likely to get?

9) Alan tosses two coins; are they more likely to land on two heads or two tails?

10) Peter spins a spinner with 3 red and 3 white faces. If he spins it twice, list all the combinations of colours he could get

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Year 4 Mathematics

Data

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Useful formulae and hints

Data tables are used to summarize findings from research or questioning, and are usually used to show information in categories

They are often useful at this level for comparing scores or preferences from two or more groups (e.g. men and women), or comparing data over time

Graphs can show

- Changes over time
- Records of certain events (for example number of students getting 60% on a test)
- Quantities at a point in time

Graphs and tables can often show the same information; visually in the case of graphs or as a summary in the case of tables. Different types of graphs are more suitable than others depending on the information to be shown

Picture graphs are a type of graph that shows information on groups of people or items, where a symbol represents a certain quantity.

For example if one * represents 5 people, then **** would represent 20 people (4 x 5)

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Exercise 1

Data Tables

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1) Tom made a table that shows how many of his classmates have each colour as their favourite

	Green	Yellow	Blue	White	Black
Girls	4	1	1	6	2
Boys	5	0	8	4	4

a) How many children in Tom's class?

b) Which colour was most popular?

c) Which colour was most popular for boys?

d) Which colours had equal numbers of children voting for it?

e) Which colour or colours had equal number of boys voting for it?

2) A group of people was asked to vote for one day as their favourite day of the week

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Men	1	3	5	10	5	6	15
Women	3	0	2	5	11	3	15

a) How many people were asked?

b) What was most people's favourite day?

c) Which day was the least favourite of women?

d) Which day had the biggest difference in the number of men and women voting for it?

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3) A man made a list of the cost of a type of blanket and a fan at different times of the year

	January	March	May	July	September	November
Blankets	$3.50	$4	$5	$6.50	$5	$4
Fans	$20	$18	$15	$10	$12	$14

a) In which of the months was the blanket the cheapest?

b) In which month was the fan dearest?

c) What was the difference in its price between a fan and a blanket in September?

d) In which month were the prices closest?

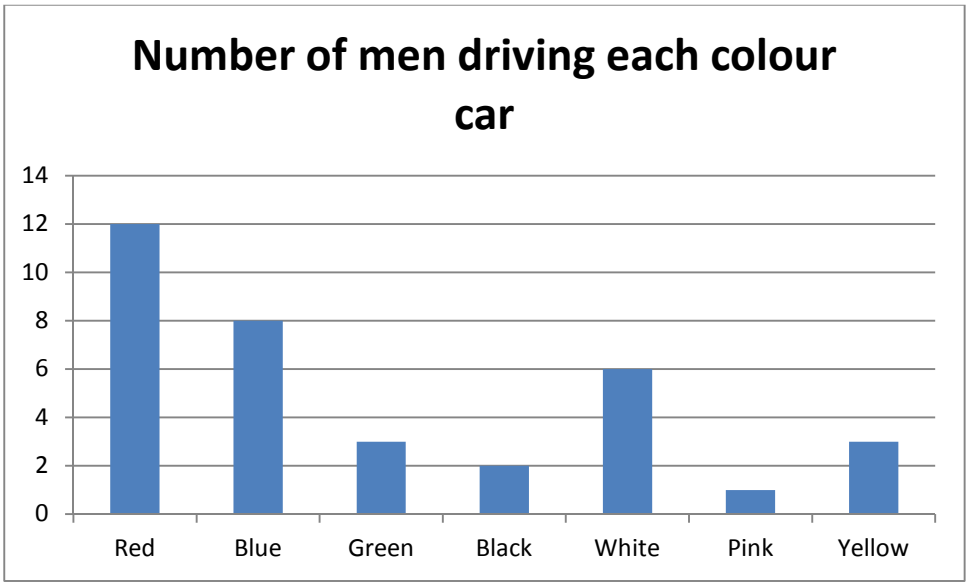
e) Explain why the prices changed so much during the year?

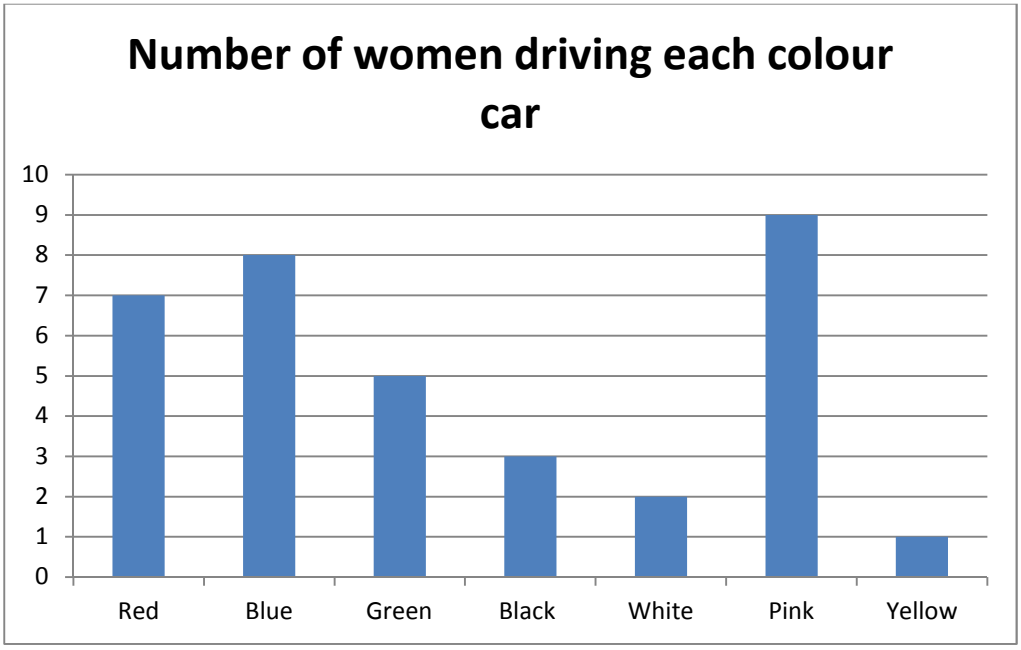
4) Show the following data in a two way table

- 100 people were surveyed as to their favourite car
- Everyone had a choice of 4 cars
- 10 men said they like Holden best
- 15 women preferred Toyota
- 5 more men than women preferred Nissan
- 10 more women than men preferred Ford
- 20 men preferred Nissan
- 12 women preferred Ford
- Equal numbers of men and women were surveyed

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5) The graphs show the number of people that own a certain colour car





a) Show the information in a two way table

b) How many people were surveyed?

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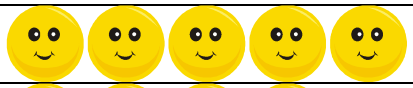
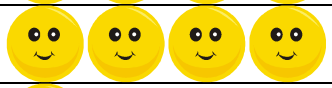
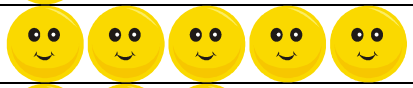
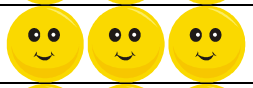
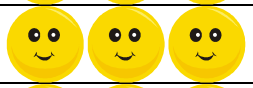
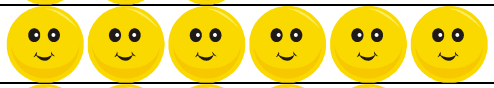
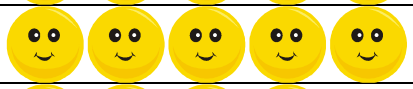
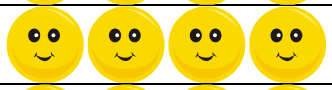
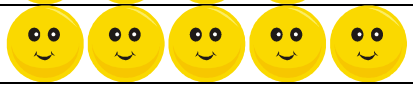
Exercise 2

Picture Graphs

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1) The picture graph below shows a sport and the number of children for whom it is their favourite

Each "face" represents 5 people

Game Number	Attendance
Football	
Rugby	
Soccer	
Basketball	
Hockey	
Swimming	
Tennis	
Golf	
Bowling	
Baseball	

a) Which sport is most popular?

b) For how many people is it their favourite?

c) For how many people is swimming their favourite sport?

d) How many people were asked?

e) Is swimming or hockey more popular?

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2) Some people were asked how many times they ate fish. The picture graph shows their answers. Each fish represents 15 days of the year

Name	Number of days eating fish
Tom	
Benny	
Jane	
Julie	
Karen	
Brian	
Richard	
Ray	
Daniel	
Craig	

a) Who eats fish the most days of the year?

b) How many days a year do they eat fish?

c) Who eats fish on the least number of days?

d) How many days do they eat fish on?

e) If someone ate fish on 50 days of the year, how could you show this on the graph? Can you think of a better way to show numbers of days that are not groups of 15?

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3) The graph below shows the number of kilos of each fruit bought in a week by a cafe. Bananas were $2.50, apples $2, oranges $3, watermelon $1.50 and strawberries $4 per kilo



a) On which fruit did the cafe spend most money?

b) What fruit did the cafe buy least of?

c) How many kilos of fruit were bought in total?

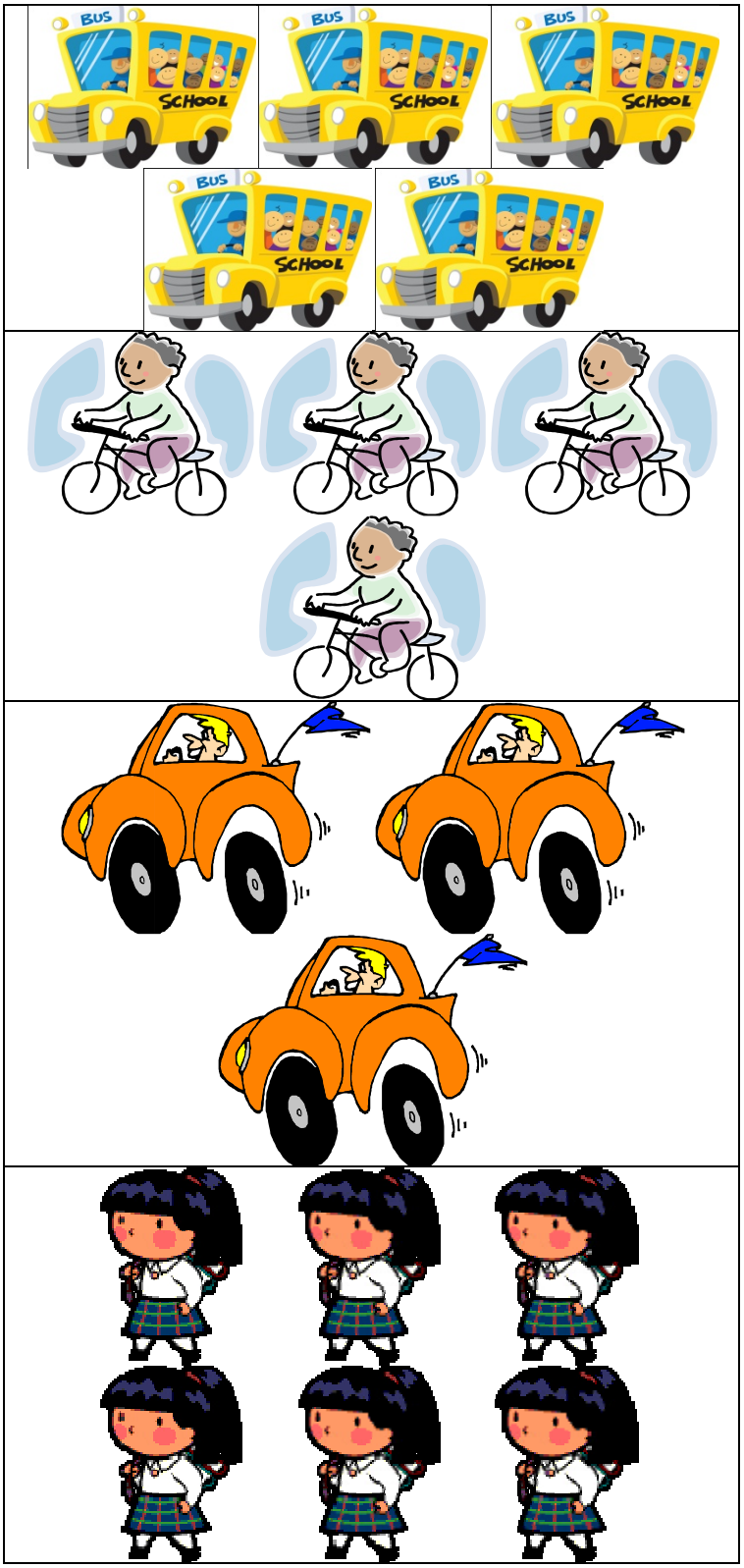
d) How much did the cafe spend on fruit in total?

4) Draw a picture graph that shows the number of people that voted for their favourite animal

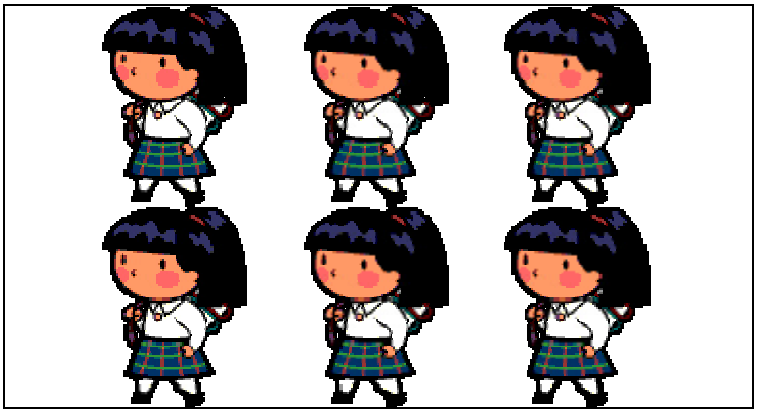
Animal	Number of men	Number of women
Dog	10	4
Cat	8	5
Rabbit	2	8
Horse	4	2
Mouse	5	0
Chicken	4	6
Lion	5	3
Tiger	3	1
Snake	1	0
Monkey	0	1

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5) The following picture graph shows the number of children that get to school in different ways. Each picture represents 10 children. Show the same information in a column graph



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Year 4 Mathematics

Space

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Useful formulae and hints

Tessellation is the process of making patterns from shapes by combining them is special ways. In this unit, we are looking to tessellate congruent shapes. That is only using the same size and type shape to tessellate.

Tessellations between these types of shapes are successful if no space is left between them, that is they fit together perfectly

There are 3 methods of treating shapes that may allow them to tessellate.

- Rotation involves revolving a shape around a fixed point on its perimeter
- Reflection involves making a mirror image of the shape
- Translation involves sliding a shape in a particular direction.

By using one or a combination of these techniques, shapes can be tested to see if they tessellate

In this unit we are looking at angles that are either right angles (also called perpendicular), and those that are more or less than right angles

An angle is made up of two line segments that meet at a point called a vertex

Some 3 dimensional shapes in this unit are

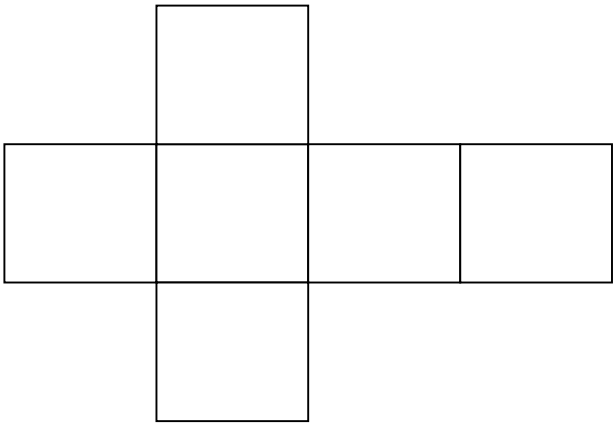
- Cylinders

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- Pyramids (square and triangular based)
- Prisms (Triangular and rectangular)
- Cones

Different views of these shapes should be able to be drawn.

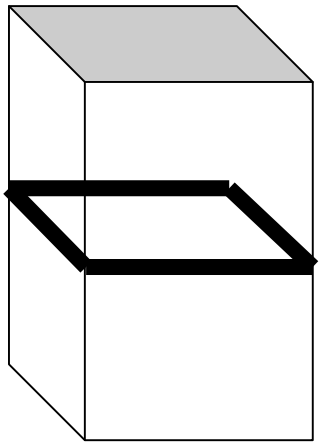
The net of a shape is the 2D (flat) representation of it. It is the model of the shape as if it were undone and flattened.



Net of a cube

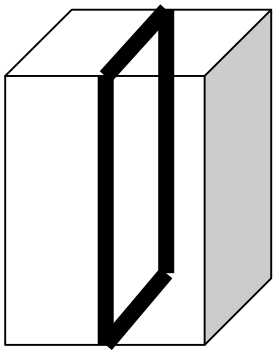
A cross section parallel to the base of a shape is the top view of a cut that goes across the shape

View of the parallel cross section of a rectangular prism



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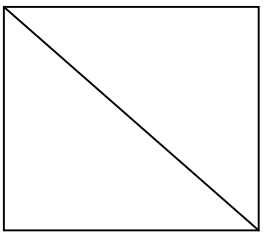
View of the perpendicular cross section of a rectangular prism



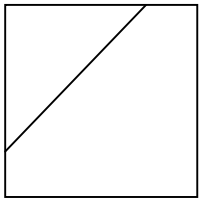
A cross section perpendicular to the base is the side view of a cut that goes down the shape

Both cross sections produce a 2 dimensional shape (e.g. a rectangle)

A line of symmetry is a line drawn from one point on the perimeter of a shape to another, such that the two halves produced are identical



Line of symmetry



Not a line of symmetry

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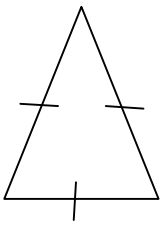
Exercise 1

Tessellations

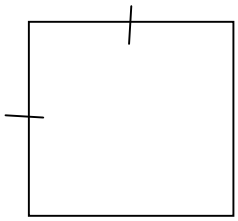
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1) Which of the following shapes tessellate?

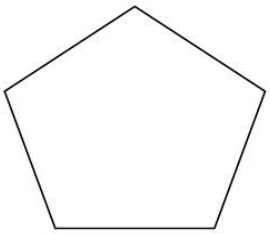
a)



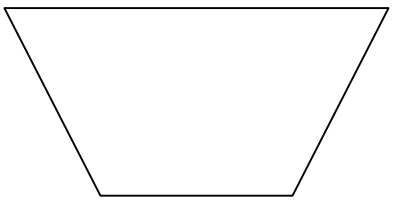
b)



c)



d)



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e)

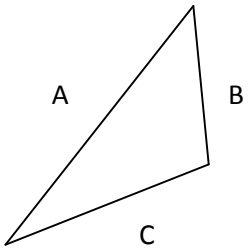


2) In the space in the table, write down how many of each shape is necessary to completely tessellate around a point

Equilateral Triangle	
Square	
Regular Pentagon	
Regular Hexagon	

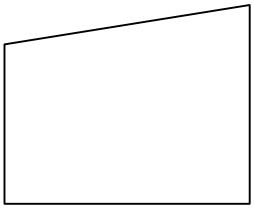
3) Explain in your own words why you need different numbers of certain shapes to be able to tessellate them

4) The side lengths of the triangle are all different. By rotating the triangle, construct a tessellation, and identify the side names in each triangle



5) Using the triangle above, form a tessellation by using a combination of rotations and a reflection?

6) By using rotations, construct a tessellation from the following quadrilateral



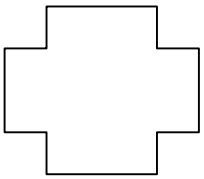
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7) By using a translation (sliding), form a tessellation from the following shape

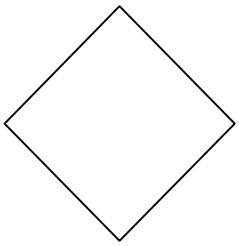


8) What technique(s) would you use to tessellate the following shapes?

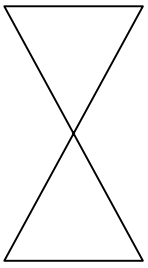
a)



b)

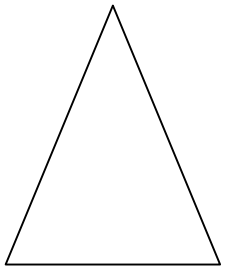


c)

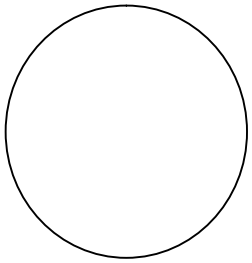


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d)



e)



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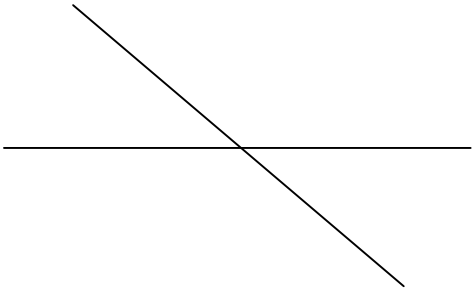
Exercise 2

Angles

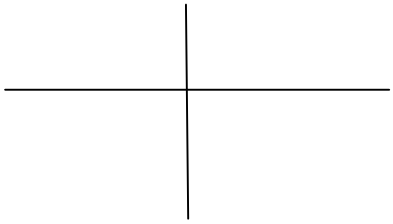
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1) Which of the following pairs of lines are perpendicular?

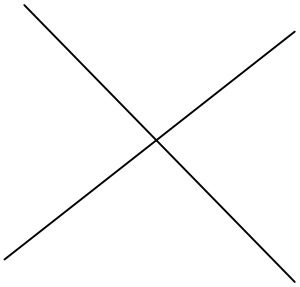
a)



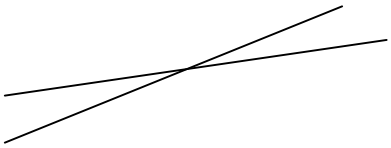
b)



c)

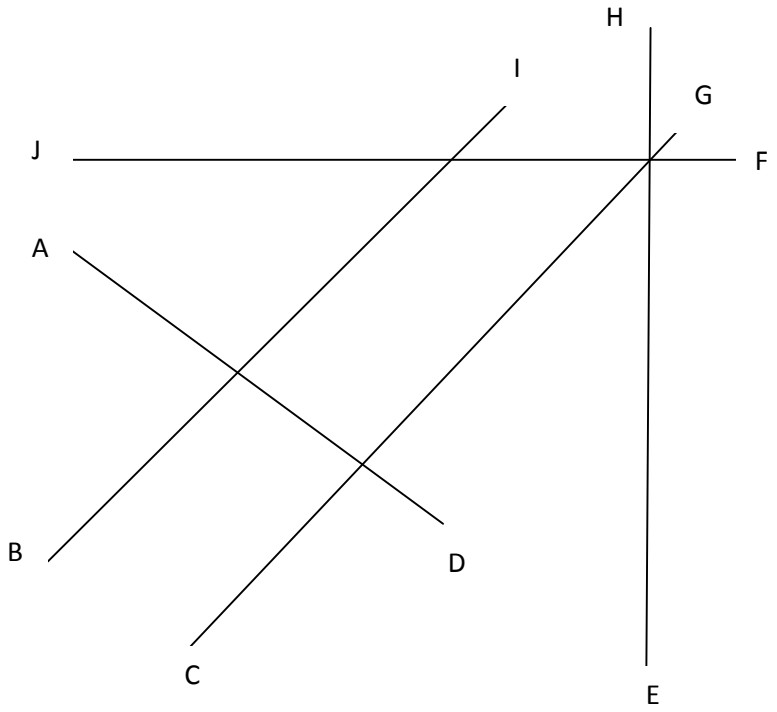


d)



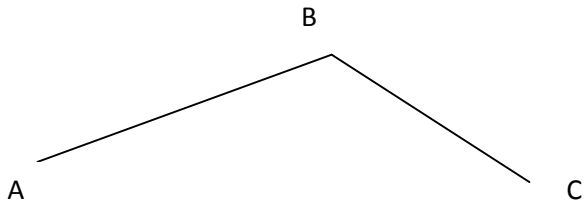
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2) In the following diagram name all the perpendicular pairs of lines

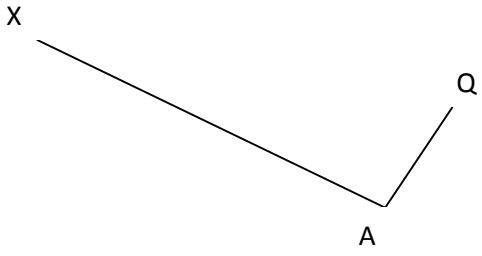


3) Which letter denotes the vertex in each of the following angles?

a)

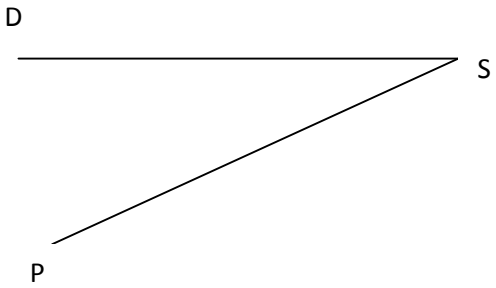


b)

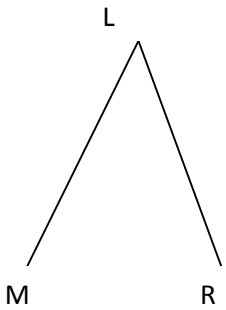


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c)



d)



e)



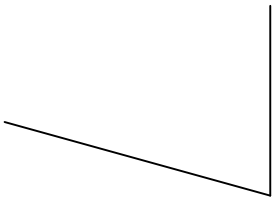
f)



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4) Describe each of the following angles as less than right-angled, more than right angled or right-angled

a)



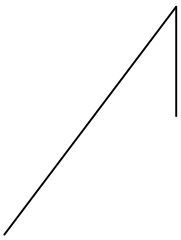
b)



c)

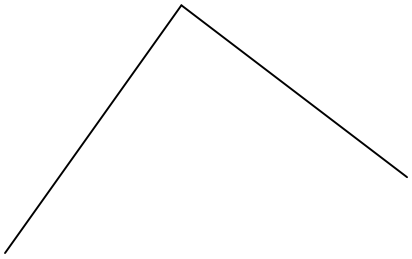


d)

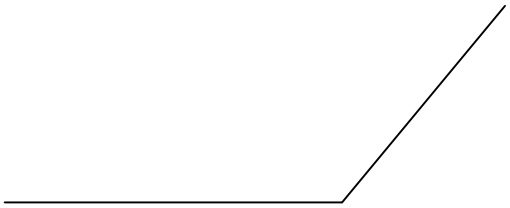


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e)

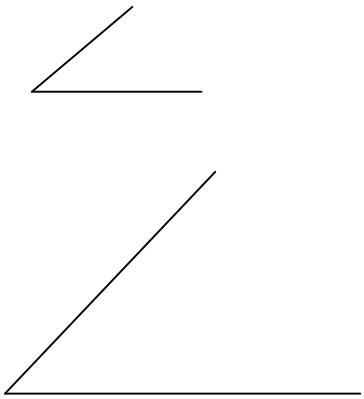


f)

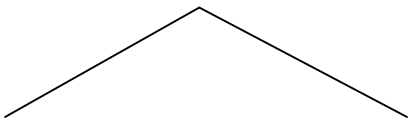


5) State whether each pair of angles are the same size

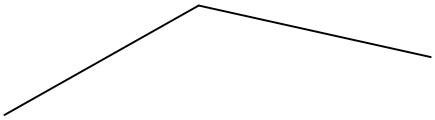
a)



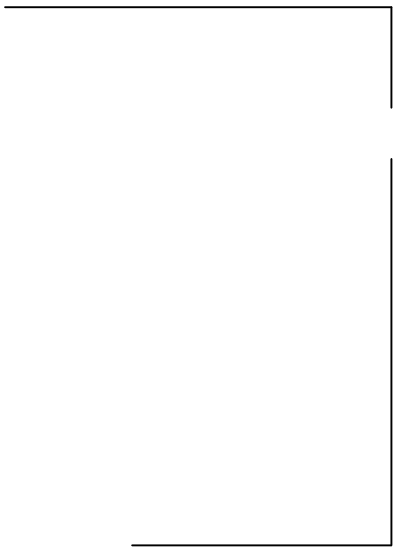
b)



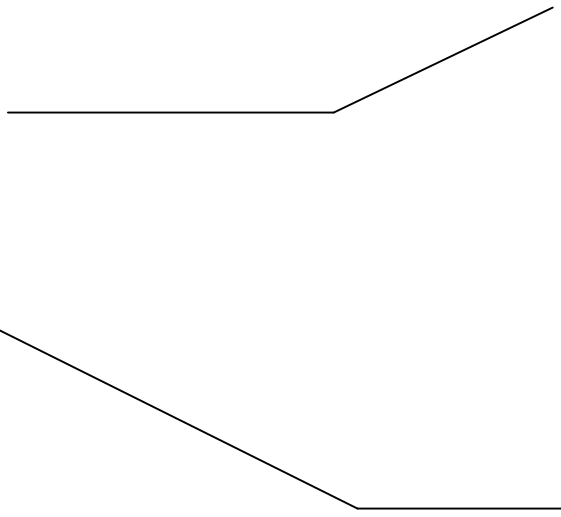
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c)



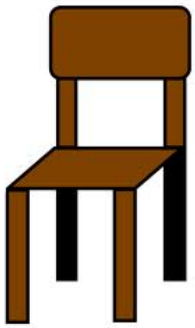
d)



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6) Identify what parts of the following objects form angles

a)



b)



c)



d)

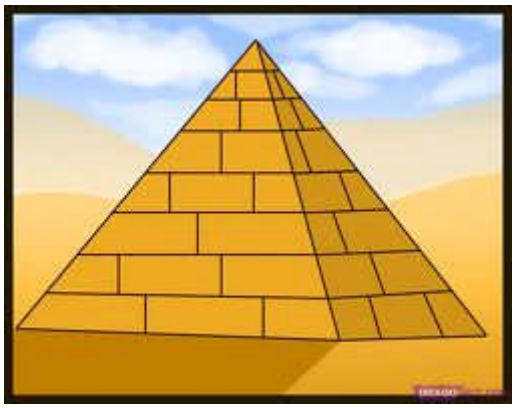


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e)



f)



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Exercise 3

2D and 3D Shapes

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1) Sketch the following shapes

a) Cylinder

b) Triangular prism

c) Triangular pyramid

d) Rectangular prism

e) Cone

f) Triangular prism

2) Sketch a cylinder from the following views

a) Side

b) Above

c) Below

3) Sketch a triangular prism from the following views

a) Side

b) Below

c) End

d) Above

4) Draw a net of the following shapes

a) Rectangular prism

b) Triangular pyramid

c) Cylinder

d) Cone

5) Draw and describe the shape formed when a cross section parallel to the base is taken of the following

a) Cylinder

b) Rectangular prism

c) Triangular pyramid

d) Cone

6) Draw and describe the shape formed when a cross section perpendicular to the base is taken of the following

a) Cone

b) Triangular prism

c) Square pyramid

d) Cylinder

7)

a) Draw the lines of symmetry of a rectangle

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b) Draw a line through a rectangle that is not a line of symmetry

8) Draw a triangle that has all sides of equal length and draw all its lines of symmetry

9) Draw a triangle that has 2 of its sides having equal length, and draw all its lines of symmetry

10) Draw a triangle that has no sides of equal length and draw all its lines of symmetry

11) Draw a square and also draw all its lines of symmetry

12) Draw a four sided shape that has no sides of equal length and draw all its lines of symmetry

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Year 4 Mathematics

Measurement

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Useful formulae and hints

Digital clocks are read as if the numbers were words. For example, the time on a digital clock reading 9:12 is read as “nine twelve” (twelve minutes past nine)

Also anything past thirty minutes can also be read as a number of minutes to the next hour. To calculate this, subtract the number of minutes showing from sixty

For example: 8:42 is read as “eight forty two” or as forty two minutes past eight

It can also be read as (60 – 42=) eighteen minutes to 9

There are 60 minutes in one hour, and 60 seconds in one minute

There are 1000 grams in 1 kg

To change grams to kg, divide by 1000

3200 g $= 3200 \div 1000 = 3.2$ kg

To change kg to grams, multiply by 1000

4.3 kg $= 4.3 \times 1000 = 4300$ grams

Example of strategy for solving word problems:

Wire is 200grams for 20 cents. How much could you buy for $5?

Answer: $5 is 25 lots of 20 cents $(500 \div 20 = 25)$

So you could buy 25 lots of 200 grams

$25 \times 200 = 5000$ grams = 5 kg of wire

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There are 100 cm in 1 metre

To convert cm to m, divide by 100

500 cm$= 500 \div 100 = 5$m

To convert m to cm, multiply by 100

7.4 m $= 7.4 \times 100 = 740$ cm

The perimeter of a shape is the distance around the outside of it (all distances must be the same unit)

If the four sides of a rectangle are 50 cm, 1 m, 50 cm, and 1 m, the perimeter is 0.5 m +1 m + 0.5 m +1 m = 3 m (or 300 cm)

The area of a rectangle is equal to the length of the rectangle multiplied by its width (all distances must be the same unit)

For the rectangle above, the area is 0.5 x 1 = 0.5 m^2, (or 50 x 100 =5000 cm^2)

The litre is the unit of volume. One litre = 1000 millilitres (mL)

The volume of a 3D shape (how much space it takes up) is measured in cm^3 (or m^3), and its capacity (how much liquid it can hold) is measured in mL (or litres)

1 cm^3 = 1 millilitre

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Exercise 1

Time

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1) Write the following times in words

a)



b)



c)



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d)



2) Write the following times in two different ways. (For example seven forty-five, quarter to 8)

a)



b)



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c)



d)



3) Convert the following to minutes

a) 1 hour

b) 2 hours

c) 1 and a half hours

d) Ten hours

e) 2 hours and fifteen minutes

f) 4 hours and ten minutes

4) Convert the following to seconds

a) One minute

b) Two minutes

c) Five minutes

d) Two and a half minutes

e) Six minutes and 20 seconds

f) 1 hour

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5) Write each of these times as they would appear on a digital clock

a) Eight thirty

b) Six forty five

c) Quarter past three

d) Half past nine

e) Ten minutes to one

f) Quarter to 8

g) Noon

6) The main movie at the theatre shows every 2 and a half hours. If it started at seven thirty, when would the next showing begin?

7) A bus goes from the city to John's street every fifteen minutes. If the last bus for the night leaves at nine o'clock, when did the second last bus leave?

8) A magazine is published every 2 weeks. If t was published on May 1st, when is the next time it would be published?

9) The American Civil War started in 1860 and went until 1865. How long did it last for?

10) It took Alan one and a half years to sail around the world. If he left on January 1st 2010, when did he return?

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Exercise 2

Mass

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1) Convert the following to grams

a) Half a kilogram

b) One quarter of a kilogram

c) One fifth of a kilogram

d) Three quarters of a kilogram

e) One third of a kilogram

2) Convert the following to kilograms

a) 500 grams

b) 750 grams

c) 250 grams

d) 100 grams

e) 1500 grams

f) 1250 grams

g) 3500 grams

3) Add the following giving your answer in kg

a) 500g + 500g

b) 700g + 700g + 600g

c) 200g + 800g

d) One and a half kg plus half a kg

e) 750g + 750g

f) One and a half kg plus one and a half kg

4) Write the following in kg

a) Four lots of 500g

b) Three lots of 500g

c) Half of 4kg

d) Five and a half kg subtract two and a half kg

e) One half of 5kg

5) Eric has a bag of marbles. Each marble weighs 200g and he has 10 of them. If John's marbles each weigh 400g, how many does he need to have the same weight of marbles as Eric?

6) Four men each carry a bag of rocks weighing 250g. How many kg do they carry between them?

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7) John has $5 and wants to buy as much paper as he can. Each 100g of paper costs 50 cents. How much paper can he buy?

8) Three books weigh 250g, 300g and 600g. How much do the books weigh together?

9) Peter has three weights: two of them weigh 400g and the other weighs 700g. Alan has two weights: one weighs 1kg and the other 500g. Who has more weight?

10) Thomas eats 500g of a 750 g steak, while his Dad leaves 100g of his. How much steak is left in total?

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Exercise 3

Length, Perimeter & Area

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1) Convert the following to metres (e.g. 1m 50cm = 1.5m)

a) 1 m 25cm

b) ½ m

c) 2 m 50cm

d) 3m 60cm

e) 2m 75cm

f) 80cm

2) Convert the following to m and cm (e.g. 1.5m = 1m 50cm)

a) 1.25m

b) 600 cm

c) 2.75m

d) 0.5m

e) 4.2m

f) 1.05m

3) Graham is 1.6m tall, while his dad is 2 metres. How much taller is Graham's dad in metres?

4) A square has side length of 1 metre, what is its area?

5) Would the area of the following be approximately equal to 1 square metre, less than 1 square metre, or more than 1 square metre?

- The floor of a kitchen
- A window
- A stamp
- A coffee table
- A lawn
- A field
- A car door

6) Describe how to calculate the perimeter of a shape

7) Calculate the perimeter of each of the following rectangles

a) Side lengths 1m and 2m

b) Side lengths 2m and 3m

c) Side lengths 5m and 4m

d) Side lengths 1.5m and 2m

e) Side lengths 1m 50cm and 2m

f) Side lengths 50cm and 1m

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8) Calculate the area of each of the following rectangles

a) Side lengths 1m and 2m

b) Side lengths 2m and 3m

c) Side lengths 5m and 4m

d) Side lengths 1.5m and 2m

e) Side lengths 1m 50cm and 2m

f) Side lengths 50cm and 1m

9) There are two pieces of wood on the ground. One has a length of 1m and a width of 4m, the other is a square piece of side length 2m. Which piece of wood has a bigger area? Which piece of wood has the bigger perimeter?

10) A man walked around a lounge room that was 3m long and 2m wide. How far did he walk??

11) The man from question 10 wishes to carpet his lounge room. How many square metres of carpet will he need?

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Exercise 4

Volume & Capacity

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1) Estimate the capacity in litres of each of the following?

- A milk carton
- A car's petrol tank
- A bath
- A large bottle of soft drink
- A swimming pool
- A kitchen sink

2) Convert the following to mL

a) 1.25 L

b) 2.6L

c) 0.75L

d) 3.9L

e) 2.24L

f) 8L

3) Convert the following to Litres

a) 4000mL

b) 2500mL

c) 1250mL

d) 4750mL

e) 10000mL

4) How much liquid is wasted if 500mL is added to a 1 litre container that already contains 750mL?

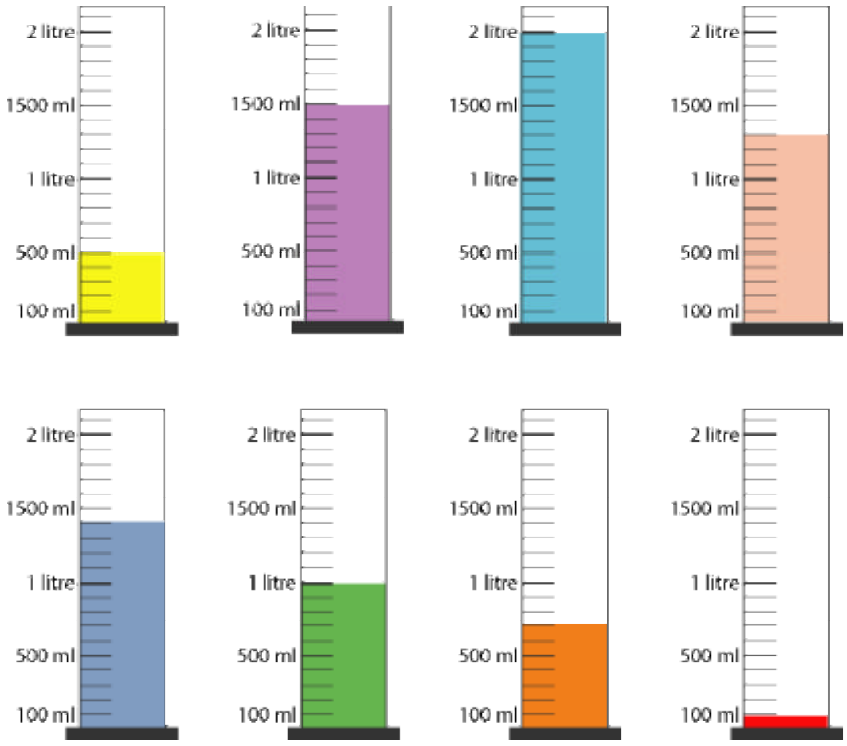
5) To fill a 2L container, how much liquid needs to be added if it currently contains 1.4 litres?

6) Bill poured 600mL of water into a bowl, Tom poured a further 500mL and Peter poured 900mL. How much water was in the container?

7) A 1 litre container is filled to the top with water. One hundred $1cm^3$ blocks are thrown into the container and water overflows as a result of this. How much water is left in the container?

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8) How much liquid is in the following cylinders?



9) Stacks of 1 cm blocks are built. How much water would they displace from a container if they were dropped in?

a) 2 rows and 3 columns

b) 4 rows and 5 columns

c) 6 rows and 3 columns

d) 3 rows and 6 columns

e) 10 rows and 10 columns

f) 30 rows and 30 columns

10) In a fridge there were five 250 mL cans of soft drink. How much soft drink was there altogether?

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